

PRACTICE EXAM 1 OF 4 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 1

Twenty questions. Domain-weighted. Nine seconds per question, or pause as you need.

QUESTIONS 20 WEIGHTING 30/26/24/20 PASS MARK 14 of 20

An application on EC2 must read secrets from AWS Secrets Manager. Developers have embedded an IAM user's access keys in the application configuration to authenticate. Security requires that no long-lived credentials exist on the instances.

What should the solutions architect do?

- ☐ A Rotate the access keys every thirty days using a Lambda function
- ☐ B Attach an IAM role to the instances granting access to the required secrets
- ☐ C Store the access keys in Systems Manager Parameter Store as a SecureString
- ☐ D Encrypt the configuration file using a KMS customer managed key

◆ PAUSE AND DECIDE

What should the solutions architect do?

- A

 Rotate the keys
The long-lived credential still exists, which is what the requirement forbids.
- C

 Keys in Parameter Store
Moves where the key is kept. The key still exists and the instance still needs a credential to fetch it.
- B

 IAM role on the instances
Removes the long-lived credential entirely and supplies temporary credentials that rotate automatically.
- D

 Encrypt the config file
Circular. Decrypting requires a credential, so the problem is unchanged.

THE SIGNAL

Long-lived keys on an instance always resolve to a role, never to better storage.

A company must ensure that S3 buckets in any of its forty-five accounts can never be made publicly readable, and that this cannot be overridden by an account administrator.

The security team wants prevention rather than detection.

Which solution meets this requirement?

- ☐ A Enable S3 Block Public Access in each account individually
- ☐ B Apply a service control policy denying the actions that make buckets public
- ☐ C Create an AWS Config rule with automatic remediation in every account
- ☐ D Use IAM Access Analyzer to report publicly accessible buckets

◆ PAUSE AND DECIDE

Which solution meets this requirement?

- A

 Block Public Access per account
An account-level setting a local administrator can turn off, and it must be maintained in forty-five places.
- C

 Config rule with remediation
Detective. The bucket becomes public before it is corrected, which the question rules out.
- B

 Service control policy
Preventive, applied above account administrators, and managed centrally across the organisation.
- D

 Access Analyzer
Reports findings. It prevents nothing.

THE SIGNAL

'Cannot be overridden by an account administrator' plus 'prevention not detection' is always an SCP.

Web servers in a private subnet must accept traffic only from an Application Load Balancer. The load balancer scales automatically and its private addresses change. A junior engineer has allowed port 443 from the VPC CIDR range.

What is the MOST secure configuration?

- ☐ A Allow port 443 from the load balancer's security group as the source
- ☐ B Allow port 443 from the public subnet CIDR ranges
- ☐ C Add a network ACL allowing port 443 from the VPC CIDR
- ☐ D Schedule a Lambda function to update the rule with current ALB IPs

◆ PAUSE AND DECIDE

What is the MOST secure configuration?

- A

 Reference the ALB security group
Grants access to exactly the load balancer, follows it as it scales, and needs no maintenance.
- B

 Public subnet CIDR
Permits anything in those subnets, which is broader than the load balancer alone.
- C

 NACL on the VPC CIDR
Even broader than the current rule, and leaves the security group unchanged.
- D

 Lambda updating IPs
Permanent maintenance plus a failure window whenever scaling outpaces the function.

THE SIGNAL

Changing addresses inside a VPC means security group referencing, every time.

A company must store customer records encrypted at rest, must be able to audit every use of the encryption key, and must be able to revoke a partner account's ability to decrypt at any moment.

The data is held in S3.

Which approach meets all three requirements?

- ☐ A SSE-S3 with default bucket encryption
- ☐ B SSE-KMS with a customer managed key and a key policy
- ☐ C Client-side encryption with keys held by the application
- ☐ D SSE-KMS with an AWS managed key

◆ PAUSE AND DECIDE

Which approach meets all three requirements?

- A

 SSE-S3
Encrypts the data but gives no key policy and no per-key audit trail.
- C

 Client-side encryption
Puts key management entirely on the company, with no managed audit trail or revocation mechanism.
- B

 SSE-KMS, customer managed key
Key policy controls who may decrypt and can be edited to revoke instantly, and CloudTrail records every key use.
- D

 AWS managed key
You cannot write a key policy for it, so cross-account access cannot be granted or revoked.

THE SIGNAL

Any requirement to audit, share or revoke key access means a customer managed key.

An engineer reports that a Lambda function cannot write to a DynamoDB table. The function's execution role has a policy allowing all DynamoDB actions on that table. The account was recently moved into a new organisational unit. Functions in other organisational units write to the table without issue.

What is the MOST likely cause?

- ☐ A The execution role is missing a trust policy for Lambda
- ☐ B A service control policy on the new OU denies the action
- ☐ C DynamoDB tables cannot be written to by Lambda functions
- ☐ D The function has run out of provisioned concurrency

◆ PAUSE AND DECIDE

What is the MOST likely cause?

- A

 Missing trust policy
The function is executing, which means the role was assumed successfully. The trust policy is working.
- B

 SCP on the new OU
Two clues: a recent move to a new OU, and identical functions elsewhere working. SCPs are evaluated before identity policies.
- C

 Lambda cannot write to DynamoDB
An invented restriction. This is one of the most common Lambda patterns.
- D

 Provisioned concurrency
Affects start-up latency, not permissions. A permission failure is not a capacity problem.

THE SIGNAL

Permissions look correct but fail, and something changed at the OU boundary. Look above the account.

Administrators need shell access to instances in private subnets. Audit requires that no inbound ports are open, no SSH keys are distributed, and every command is recorded. The company wants the least operational overhead.

Which solution meets these requirements?

- ☐ A A bastion host in a public subnet with key rotation
- ☐ B AWS Systems Manager Session Manager with logging enabled
- ☐ C A Site-to-Site VPN with SSH access from the corporate network
- ☐ D EC2 Instance Connect from the console

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A

 Bastion with key rotation
Keeps an open port on the target and distributed keys, plus the rotation burden.
- C

 VPN plus SSH
Gets administrators onto the network but still requires an open SSH port and a key on each instance.
- B

 Session Manager with logging
No inbound ports, no keys, and full session recording to S3 or CloudWatch Logs.
- D

 EC2 Instance Connect
Still requires port 22 open to the Instance Connect service range, so it fails the no-inbound-ports requirement.

THE SIGNAL

No ports, no keys, full audit logging is the exact Session Manager feature set.

An application accepts orders through a web tier and writes them to a processing service that can handle four hundred orders per second. Marketing campaigns cause bursts of about eight thousand orders per second for two to three minutes.

Orders may be confirmed by email a few minutes after submission.

Which change best handles the bursts?

- ☐ A Place an SQS queue between the web tier and the processing service
- ☐ B Publish orders to an SNS topic subscribed by the processing service
- ☐ C Scale the processing service to handle eight thousand per second continuously
- ☐ D Add a CloudFront distribution in front of the web tier

◆ PAUSE AND DECIDE

Which change best handles the bursts?

- A

SQS queue
Accepts orders instantly and drains at the rate the processor sustains. Delayed confirmation is explicitly acceptable.
- B

SNS topic
Pushes immediately with no storage, so it forwards the burst to the same overwhelmed service.
- C

Scale to peak permanently
Pays for twenty times the capacity all year to serve a few minutes of it.
- D

CloudFront
Caches content. Order submissions are unique writes and cannot be cached.

THE SIGNAL

Short extreme burst plus tolerance for delayed completion is the queue signature.

A business requires that a Region-wide failure causes at most fifteen minutes of data loss and at most three hours of downtime for its RDS-backed application.

Cost is a stated concern and the finance team has rejected a continuously running duplicate environment.

Which DR strategy is MOST appropriate?

- ☐ A Backup and restore from nightly cross-Region snapshots
- ☐ B Pilot light with a cross-Region read replica and infrastructure as code
- ☐ C Warm standby running a scaled-down full environment
- ☐ D Multi-site active-active across two Regions

◆ PAUSE AND DECIDE

Which DR strategy is MOST appropriate?

- A

Nightly snapshots
Fails the RPO. A nightly snapshot risks losing up to twenty-four hours, not fifteen minutes.
- B

Pilot light
Continuous replication keeps data loss to seconds, and three hours is ample to promote and provision the rest.
- C

Warm standby
Meets both objectives and runs a duplicate environment, which finance has explicitly rejected.
- D

Active-active
The most expensive option, buying near-zero recovery when three hours was acceptable.

THE SIGNAL

Eliminate on RPO first, then RTO, then take the cheapest survivor.

An Auto Scaling group of ten instances sits behind an Application Load Balancer. All ten report healthy, but users intermittently receive errors. One instance has a hung application process while the operating system runs normally.

The group uses the default health check configuration.

What change will cause the instance to be replaced?

- ☐ A Set the health check type to ELB
- ☐ B Reduce the health check grace period
- ☐ C Increase the desired capacity
- ☐ D Enable detailed monitoring

◆ PAUSE AND DECIDE

What change will cause the instance to be replaced?

- A

 Health check type ELB
The load balancer tests the application, so a hung process fails the check and Auto Scaling replaces the instance.
- B

 Shorter grace period
Governs how long a new instance is given before checks apply. Irrelevant to a long-running instance.
- C

 More instances
Dilutes the error rate. The broken instance stays in rotation.
- D

 Detailed monitoring
Improves metric resolution without changing what counts as healthy.

THE SIGNAL

Running but not serving is always the EC2 versus ELB health check distinction.

Four independent teams must each receive every order event. One team's consumer is regularly offline for maintenance windows of up to two hours, and no event may be lost. New consuming teams are expected next year.

Which architecture meets these requirements?

- ☐ A One SQS queue that all four teams read from
- ☐ B An SNS topic with the four consumers subscribed directly
- ☐ C An SNS topic with four SQS queues subscribed, one per team
- ☐ D The producer calling each consumer's API in sequence

◆ PAUSE AND DECIDE

Which architecture meets these requirements?

- A

 Shared SQS queue
A message is consumed once, so the first reader denies it to the other three.
- B

 SNS direct subscriptions
SNS does not store. Events published during the two-hour outage are lost.
- C

 SNS fan-out to four queues
Each consumer gets a durable buffer, offline consumers lose nothing, and new teams just add a subscription.
- D

 Sequential API calls
Tightly coupled. One offline consumer blocks the producer, and each new team needs a producer change.

THE SIGNAL

'No event may be lost while a consumer is down' is what upgrades SNS alone to SNS plus SQS.

A company must retain audit logs in S3 for seven years in a form that cannot be altered or deleted by anyone, including the account root user.

A separate bucket holds working documents that staff edit daily, and the team wants protection against accidental overwrites.

Which pairing is correct?

- | | | | |
|----------------------------|--|----------------------------|--|
| ☐ A | Object Lock compliance mode for the logs; versioning for the documents | ☐ B | Object Lock governance mode for the logs; versioning for the documents |
| ☐ C | Versioning for the logs; Object Lock compliance mode for the documents | ☐ D | A bucket policy denying deletes for both |

◆ PAUSE AND DECIDE

Which pairing is correct?

- A

 Compliance mode, then versioning
Compliance mode admits no override including root, and versioning makes overwrites recoverable without blocking daily editing.
- B

 Governance mode for the logs
Governance mode lets privileged users override retention, which fails the 'including root user' requirement.
- C

 Reversed
Versioning cannot prevent deletion, and compliance mode on working documents would stop staff editing them.
- D

 Bucket policy for both
A policy can be edited by whoever holds the permission, so it guarantees nothing.

THE SIGNAL

'Including the root user' selects compliance mode specifically over governance mode.

A batch job processes files for approximately fifty minutes per run and executes four times daily. It currently runs on an always-on EC2 instance.

The team wants to remove server management and pay only for processing time.

Which solution is MOST appropriate?

- ☐ A Rewrite the job as a Lambda function triggered by EventBridge
- ☐ B Run the job as a scheduled AWS Fargate task
- ☐ C Split the job across chained Lambda functions
- ☐ D Move the job to a smaller always-on instance

◆ PAUSE AND DECIDE

Which solution is MOST appropriate?

- A

 Lambda
Fifty minutes exceeds the fifteen-minute hard limit. Impossible regardless of how well it otherwise fits.
- B

 Scheduled Fargate task
Serverless with no runtime limit. Nothing runs between executions, so you pay only for the four runs.
- C

 Chained Lambdas
Adds checkpointing complexity to design around a limit Fargate does not have.
- D

 Smaller always-on instance
Keeps the server to manage and keeps paying for twenty-two idle hours a day.

THE SIGNAL

Serverless is not a synonym for Lambda. Check runtime against fifteen minutes first.

Twelve EC2 instances running a Linux content system must all read and write the same set of files, with changes visible immediately across every instance.

The instances span three Availability Zones.

Which storage service should be used?

- | | |
|---|---|
| ☐ A Amazon EBS volumes with a synchronisation script | ☐ B Amazon EFS |
| ☐ C Amazon FSx for Windows File Server | ☐ D Instance store volumes |

◆ PAUSE AND DECIDE

Which storage service should be used?

A EBS with sync script

Synchronisation always has a window where instances disagree, and EBS attaches to one instance.

C FSx for Windows

SMB file shares for Windows. This is a Linux workload.

B Amazon EFS

A shared file system mounted by every instance across all three zones, with writes immediately visible everywhere.

D Instance store

Local to each instance, not shared, and lost when an instance stops.

THE SIGNAL

Shared across many Linux instances, across AZs, is always EFS.

A DynamoDB table is throttling write requests. All items use a status field with four possible values as the partition key.

Provisioned write capacity is more than double the measured request rate.

What is the cause and fix?

- ☐ A Insufficient capacity; increase provisioned write units
- ☐ B A hot partition; change to a high-cardinality partition key
- ☐ C Missing cache; add a DAX cluster
- ☐ D Missing index; add a global secondary index

◆ PAUSE AND DECIDE

What is the cause and fix?

- A

 More capacity
Capacity is already double the request rate. The bottleneck is one partition, not the total.
- B

 High-cardinality key
Four possible key values means writes concentrate on very few partitions. Spreading them makes the table's capacity usable.
- C

 DAX cluster
DAX is a read cache. This is write throttling.
- D

 Global secondary index
Adds an access pattern and consumes additional write capacity. It would make throttling worse.

THE SIGNAL

Throttling alongside spare capacity always means a hot partition.

Streaming sensor readings must be delivered to S3 for later analysis. A delay of one to two minutes is acceptable. The team has no capacity to write or operate stream-processing code.

The data must be converted to Parquet on the way in.

Which service should be used?

- ☐ A Kinesis Data Streams with a custom consumer
- ☐ B Kinesis Data Firehose
- ☐ C Amazon MSK with Kafka Connect
- ☐ D Amazon SQS with a Lambda consumer

◆ PAUSE AND DECIDE

Which service should be used?

- A

 Data Streams with custom consumer
Requires exactly the consumer code and shard management the team says it cannot take on.
- B

 Firehose
Fully managed delivery, native Parquet conversion, and buffering that matches the one to two minute tolerance.
- C

 MSK with Kafka Connect
The most operational work of any option, and Kafka is not a stated requirement.
- D

 SQS with Lambda
You would write the batching, buffering and format conversion yourself.

THE SIGNAL

'A minute or two is fine' plus 'no code' is the Firehose signature.

A global audience downloads the same catalogue of PDF documents repeatedly over HTTPS. Users far from the origin Region report slow downloads, and outbound data transfer dominates the bill.

The origin is an S3 bucket that must not become publicly accessible.

Which solution addresses both problems?

- | | |
|--|---|
| ☐ A AWS Global Accelerator in front of the bucket | ☐ B CloudFront with Origin Access Control |
| ☐ C S3 Transfer Acceleration | ☐ D Cross-Region replication to five Regions |

◆ PAUSE AND DECIDE

Which solution addresses both problems?

- A

 Global Accelerator
Improves the path but caches nothing, so every download still leaves the origin and the transfer bill is unchanged.
- B

 CloudFront with OAC
Caches near users to fix latency, cuts origin egress cost, and reads the private bucket without making it public.
- C

 Transfer Acceleration
Accelerates uploads into S3. These are downloads.
- D

 Replicate to five Regions
Five full copies plus replication transfer, which is far more expensive than caching.

THE SIGNAL

Repeatedly downloaded content over HTTPS is cacheable, so CloudFront beats Global Accelerator here.

A rendering workload runs on one hundred and fifty instances for six hours each night. Jobs checkpoint every two minutes and any interrupted work is automatically reassigned.

The company wants the largest possible saving.

Which purchasing option should be used?

- | | |
|--|--|
| ☐ A Three-year Reserved Instances | ☐ B Spot Instances |
| ☐ C A one-year Compute Savings Plan | ☐ D On-Demand with Auto Scaling |

◆ PAUSE AND DECIDE

Which purchasing option should be used?

A Reserved Instances

Reservations bill every hour of the term. Reserving 150 instances for six hours a night wastes three quarters of what you pay for.

C Savings Plan

Better than reservations here, but still discounts less than Spot for genuinely interruptible work.

B Spot

Checkpointed and automatically reassigned work is exactly Spot's case, and it gives the deepest discount available.

D On-Demand

No discount at all.

THE SIGNAL

'Checkpoints' and 'work is reassigned' are the exam telling you interruption is acceptable.

Instances in private subnets transfer around sixty terabytes per month to and from S3 in the same Region, all routed through a NAT gateway.

The NAT data processing charge is the largest line on the bill. Instances must remain private.

What is the MOST cost-effective fix?

- ☐ A A gateway VPC endpoint for S3
- ☐ B An interface VPC endpoint for S3
- ☐ C Additional NAT gateways per Availability Zone
- ☐ D Moving the instances to public subnets

◆ PAUSE AND DECIDE

What is the MOST cost-effective fix?

- A** Gateway endpoint
S3 traffic bypasses the NAT gateway completely, and gateway endpoints carry no hourly or per-GB charge at all.
- B** Interface endpoint
Also removes NAT processing but bills hourly and per GB. At sixty terabytes that is a large avoidable cost.
- C** More NAT gateways
Processing is charged per GB, so extra gateways add hourly cost without reducing it.
- D** Public subnets
Violates the requirement that instances remain private.

THE SIGNAL

Gateway endpoints are free and cover S3 and DynamoDB. Interface endpoints are charged.

Application logs in S3 Standard are queried heavily for fourteen days, occasionally until day ninety, and then retained for five years for compliance without being read.

The bucket holds six hundred terabytes.

Which approach is MOST cost-effective?

- ☐ A Lifecycle to Standard-IA at 14 days and Glacier Deep Archive at 90 days
- ☐ B Enable Intelligent-Tiering on the bucket
- ☐ C Lifecycle to Glacier Deep Archive at 14 days
- ☐ D Keep everything in S3 Standard

◆ PAUSE AND DECIDE

Which approach is MOST cost-effective?

- A

 IA at 14, Deep Archive at 90
Maps the three described phases directly onto the right classes at the right ages.
- B

 Intelligent-Tiering
For unknown patterns. Here the pattern is stated precisely, so you would pay monitoring fees for information you have.
- C

 Deep Archive at 14 days
Days 14 to 90 need occasional access, and Deep Archive retrieval takes twelve hours.
- D

 All Standard
Keeps five years of unread data in the most expensive class.

THE SIGNAL

Pattern described by age means a lifecycle policy. Pattern unknown means Intelligent-Tiering.

EC2 spending has grown twenty-five percent in a year with no traffic increase. The team suspects instances were over-sized during a migration and never revisited. There are six hundred instances across nine accounts.

Leadership has asked what to do first.

What should the solutions architect recommend FIRST?

- ☐ A Purchase three-year Savings Plans covering current usage
- ☐ B Use AWS Compute Optimizer to identify over-provisioned instances and right-size
- ☐ C Migrate all workloads to Spot Instances
- ☐ D Configure AWS Budgets alerts on the accounts

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

- A

 Buy Savings Plans
Commits three years of spending on capacity everyone suspects is too large, locking in the waste at a discount.
- B

 Compute Optimizer, then right-size
Uses real utilisation data across accounts to find the waste. Reduce consumption before discounting what remains.
- C

 Everything to Spot
Production workloads cannot all tolerate interruption, and the instances would still be over-sized.
- D

 Budgets alerts
Reports spending. It reduces nothing.

THE SIGNAL

Right-size first, then commit. The word FIRST is doing the work in this question.

Scoring Practice Exam 1

Count your correct answers, then work out where to go next.

- ◆ **17 to 20.** Comfortably above the pass mark. Move on to Practice Exam 2 after a day or two.
- ◆ **14 to 16.** Around the pass line. Rewatch the domain your misses cluster in, then sit Exam 2.
- ◆ **10 to 13.** Rewatch the two weakest domains fully before attempting Exam 2.
- ◆ **Below 10.** Work through modules 1 to 17 again in order. This is normal on a first attempt.

ON THE EXAM

Which domain your misses cluster in matters far more than the total. Questions 1 to 6 were security, 7 to 11 resilience, 12 to 16 performance, 17 to 20 cost.